

Image Generators

Source of truth for the two hand-built diagrams in this repo's documentation. Each diagram is generated from a Python script that emits SVG directly — no drawing tool, no binary source file. Editing a diagram means editing a data structure at the top of its script and re-running it.

Script	Emits	Consumed by
<code>generate_isotope_diagram.py</code>	<code>isotope-diagram.svg / .png</code>	root README §1
<code>generate_tier_ladder_diagram.py</code>	<code>tier-ladder-diagram.svg / .png</code>	root README §1

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1.0 Prerequisites

- **uv** — this directory is a self-contained uv project (`pyproject.toml`, `uv.lock`, Python pinned to 3.13 in `.python-version`). `uv run` syncs `.venv/` on first use; there is nothing to `pip install`.
- **cairo** — `cairosvg` is a binding, not an implementation. The native library must be present for PNG output: `brew install cairo`. SVG generation needs nothing beyond Python.

2.0 Running

From the repo root:

```
make generate_isotope_diagram
```

Or directly, from this directory:

```
DYLD_FALLBACK_LIBRARY_PATH=/opt/homebrew/lib uv run python
generate_isotope_diagram.py
DYLD_FALLBACK_LIBRARY_PATH=/opt/homebrew/lib uv run python
generate_tier_ladder_diagram.py
```

2.1 Why DYLD_FALLBACK_LIBRARY_PATH

`cairocffi` resolves `cairo` at import time by calling `dlopen` with a bare library name (`libcairo.2.dylib`). macOS searches the dyld shared cache, `/usr/lib`, and `/usr/local/lib` — but never `/opt/homebrew/lib`, where Homebrew installs on Apple Silicon. The variable adds that directory to dyld's last-resort search path so the name resolves. It is macOS/Homebrew-specific; on Intel Macs the prefix is `/usr/local`, and on Linux the variable is not needed at all.

Note that it must be set as a command prefix. System Integrity Protection strips all `DYLD_*` variables when a protected binary such as `/bin/zsh` is exec'd, so exporting it from a shell profile is unreliable.

3.0 Flags

Both scripts accept:

- `--out-dir DIR` — where to write output (default: current directory; created if absent)
- `--no-png` — emit SVG only, skipping `cairo` entirely

`generate_tier_ladder_diagram.py` additionally accepts `--scale N` (default `2.0`), which multiplies the 680pt viewbox width to set PNG resolution. The isotope diagram has no scale flag; its PNG width is fixed at the 2040pt document width.

4.0 Output location

Both scripts default `--out-dir` to the **current working directory**, so a bare run from here leaves the files here — while the root README embeds `docs/image_generators/isotope-diagram.png` and `docs/image_generators/tier-ladder-diagram.png`. To regenerate the images the documentation actually displays, target the parent directory:

```
DYLD_FALLBACK_LIBRARY_PATH=/opt/homebrew/lib uv run python
generate_isotope_diagram.py --out-dir ..
```

Only the PNGs are committed; the SVGs are intermediates.

5.0 Editing a diagram

Both scripts are data-driven — layout is computed, not hand-placed, so adding a row or a tier reflows everything below it.

- `generate_isotope_diagram.py` — colors live in `PALETTE`, content in the `*_ROWS` lists and `OPTIONAL_BOX`, and column x-origins and box geometry in the `LAYOUT` block.
- `generate_tier_ladder_diagram.py` — everything that changes between revisions is in `TIERS` and `DIVIDER`: counts, blurbs, example questions, and tier order (first in the list renders as the top rung). The legend total is summed from `TIERS`, so re-tiering a question cannot leave a stale count behind.

6.0 Troubleshooting

A wall of `OSError: no library called "cairo-2" was found` — `cairo` is missing or unreachable. Install it (`brew install cairo`) and set `DYLD_FALLBACK_LIBRARY_PATH` as above.

`generate_tier_ladder_diagram.py` fails this way because it catches only `ImportError`; `generate_isotope_diagram.py` catches `OSError` too and degrades to SVG-only.

cairosvg not installed; skipping PNG — usually inaccurate. `cairosvg` is a declared dependency and `uv` installs it; the message also fires when the package imports fine but the native `cairo` library cannot be loaded. Check the `DYLD_FALLBACK_LIBRARY_PATH` prefix before reaching for `pip`.

VIRTUAL_ENV ... does not match the project environment path — harmless. `uv` is noting that an unrelated active `virtualenv` is being ignored in favor of this project's `.venv/`, which is what you want.